

The background of the slide features a light gray gradient. It is decorated with numerous realistic water droplets of various sizes, some clustered in the top left and others scattered towards the bottom right. A faint, semi-transparent globe is centered in the upper half of the image, behind the main text.

HABOT HIRING PROJECT

JUNIOR GOOGLE & DEVOPS ENGINEER

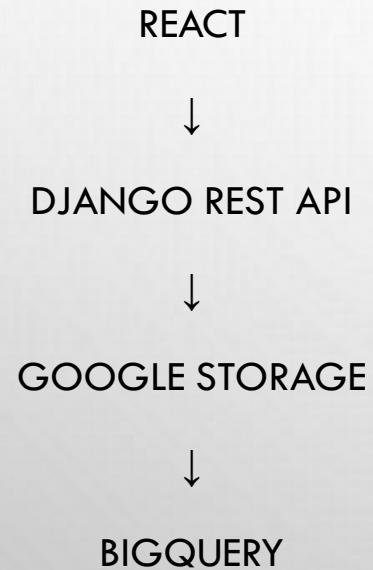
INTRODUCTION

NAME: KAVITHA KUMARI

EMAIL: kavithakumari351@gmail.com

PROJECT REPOSITORY: <https://github.com/kavitha351/habot-assesment.git>

ABOUT PROJECT



Designed a secure Google Google infrastructure using Terraform, implemented a fail-closed CI/CD pipeline using Github Actions, and created strict Django REST framework validation for student onboarding data.

TERRAFORM

TERRAFORM



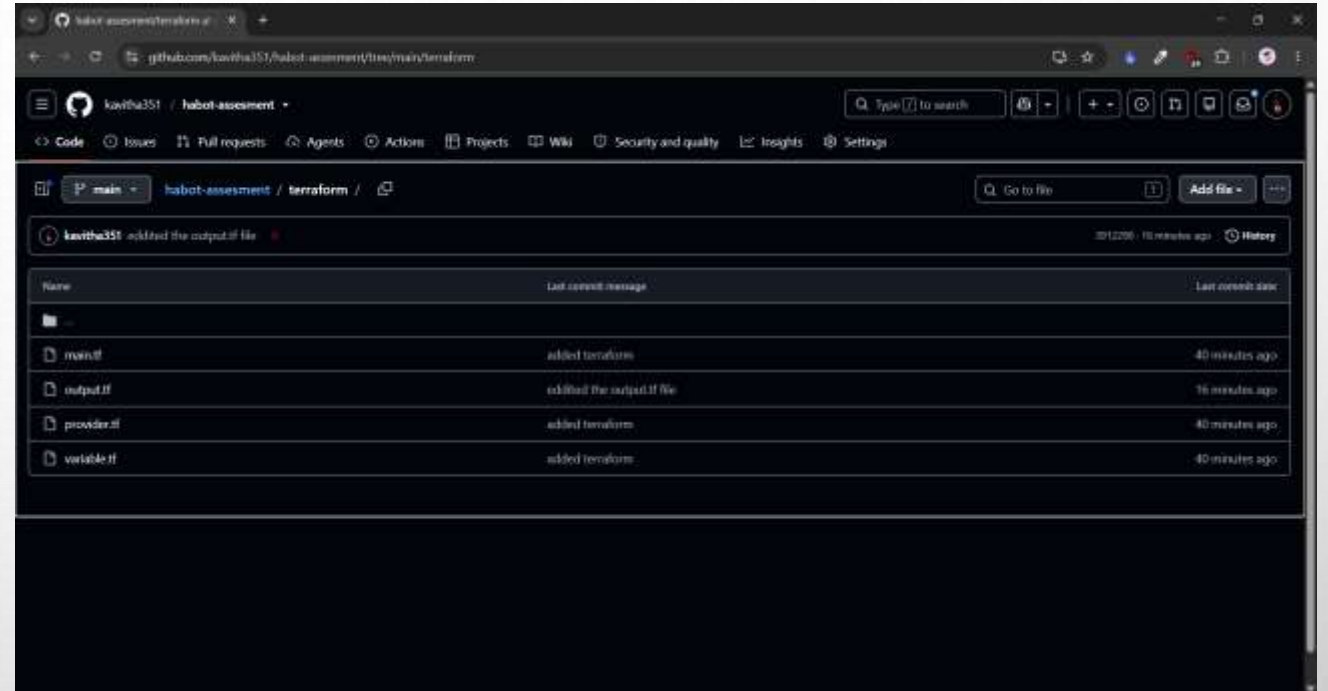
GOOGLE CLOUD STORAGE
(RAW LANDING BUCKET)



BIGQUERY DATASET



IAM POLICIES
(LEAST PRIVILEGE)



SECURITY FEATURES

☐ UNIFORM BUCKET LEVEL ACCESS

 PUBLIC ACCESS PREVENTION

 IAM LEAST PRIVILEGE

 BIGQUERY READ-ONLY ACCESS

 TERRAFORM VALIDATION

CI/CD PIPELINE

DEVELOPER



GITHUB



GITHUB ACTIONS



TERRAFORM FORMAT



TERRAFORM VALIDATE



PYTHON LINT



SECRET SCAN



PASS / FAIL

WORKFLOW

Workflow runs · kavitha351/habot-assement

github.com/kavitha351/habot-assement/actions

kavitha351 / habot-assement

Type to search

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Actions [New workflow](#)

All workflows

Habot CI Pipeline

Management

- Caches
- Attestations
- Runners
- Usage metrics
- Performance metrics

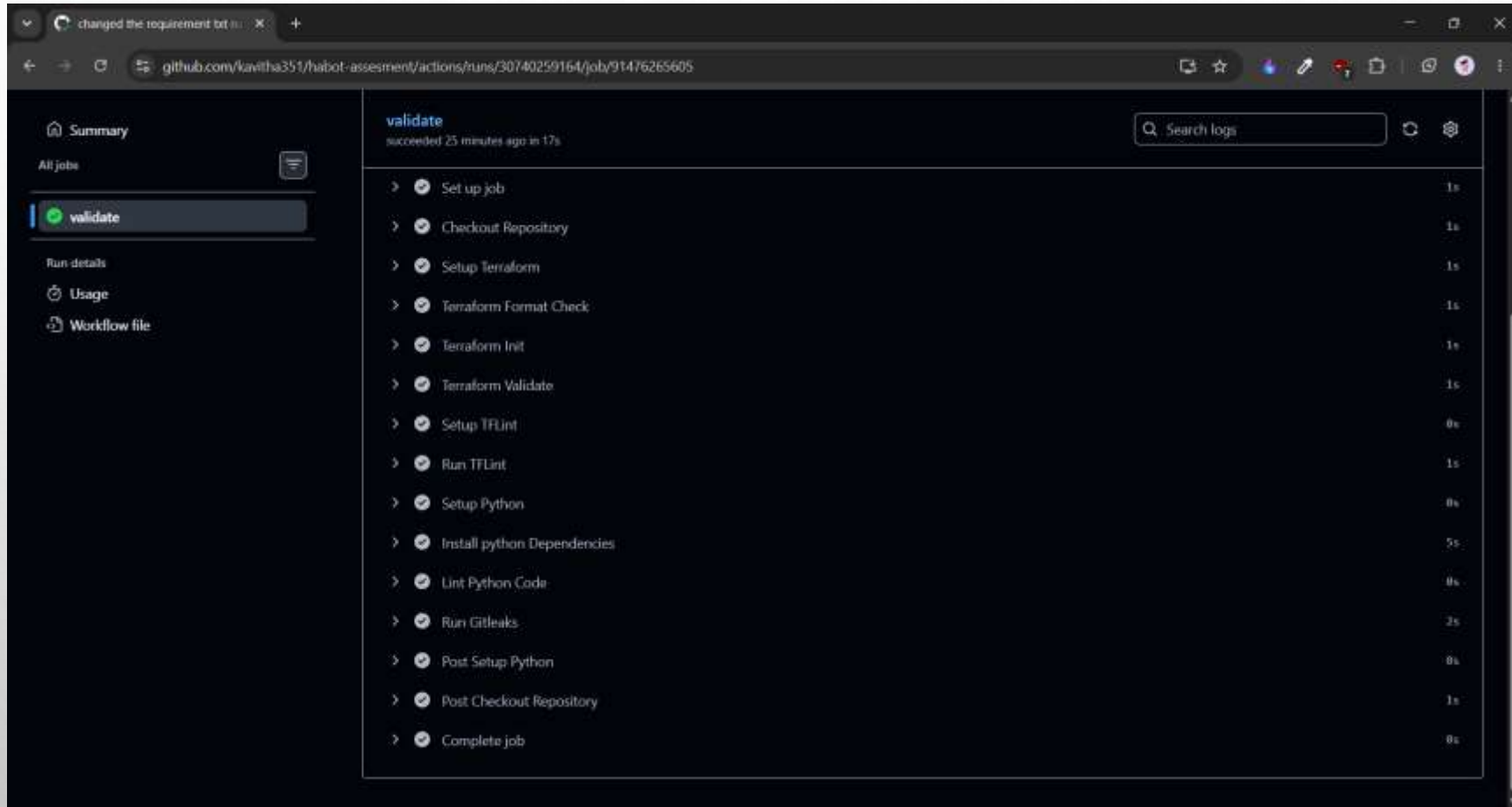
All workflows [Filter workflow runs](#)

Showing runs from all workflows

4 workflow runs

	Workflow	Event	Status	Branch	Actor
✓	changed the requirement txt name	main	23 minutes ago 27s		...
✗	editted the output.tf file	main	25 minutes ago 18s		...
✗	testing the workflow	main	29 minutes ago 12s		...
✗	added terraform	main	47 minutes ago Failure		...

WORKFLOW STEPS



The screenshot shows a GitHub Actions workflow run for the 'validate' job. The interface is dark-themed. On the left, a sidebar contains a 'Summary' section with a list of jobs: 'All jobs' (expanded) and 'Run details'. The 'validate' job is highlighted with a green checkmark. Below the sidebar, the main content area displays the 'validate' job status as 'succeeded 25 minutes ago in 17s'. A search bar labeled 'Search logs' is located in the top right of the main content area. The workflow steps are listed in a table with expandable arrows, checkmarks, step names, and durations.

Step	Duration
> ✓ Set up job	1s
> ✓ Checkout Repository	1s
> ✓ Setup Terraform	1s
> ✓ Terraform Format Check	1s
> ✓ Terraform Init	1s
> ✓ Terraform Validate	1s
> ✓ Setup TFLint	0s
> ✓ Run TFLint	1s
> ✓ Setup Python	0s
> ✓ Install python Dependencies	5s
> ✓ Lint Python Code	0s
> ✓ Run Gitleaks	2s
> ✓ Post Setup Python	0s
> ✓ Post Checkout Repository	1s
> ✓ Complete job	0s

WORKFLOW RESULT

SUCCESSFUL EXECUTION OF THE CI/CD PIPELINE AFTER ALL VALIDATION CHECKS PASSED.

The screenshot shows a web browser window displaying the GitHub Actions workflow run summary for a job named 'validate'. The URL in the address bar is `github.com/kavitha351/habot-asesment/actions/runs/30740259164`. The left sidebar contains navigation links: 'Summary' (selected), 'All jobs', 'Run details', 'Usage', and 'Workflow file'. The main content area shows the 'validate summary' section with a large green checkmark and the text 'No leaks detected'. Below this, there is an 'Annotations' section with a warning icon and a message about Node.js 20 being deprecated. The 'Artifacts' section shows a table with one artifact named 'gitleaks-results.sarif'.

changed the requirement.txt file

github.com/kavitha351/habot-asesment/actions/runs/30740259164

Summary

All jobs

validate

Run details

Usage

Workflow file

validate summary

No leaks detected ✓

[Job summary generated at run-time](#)

Annotations

1 warning

⚠ validate

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python@v5, gitleaks-...

Show more

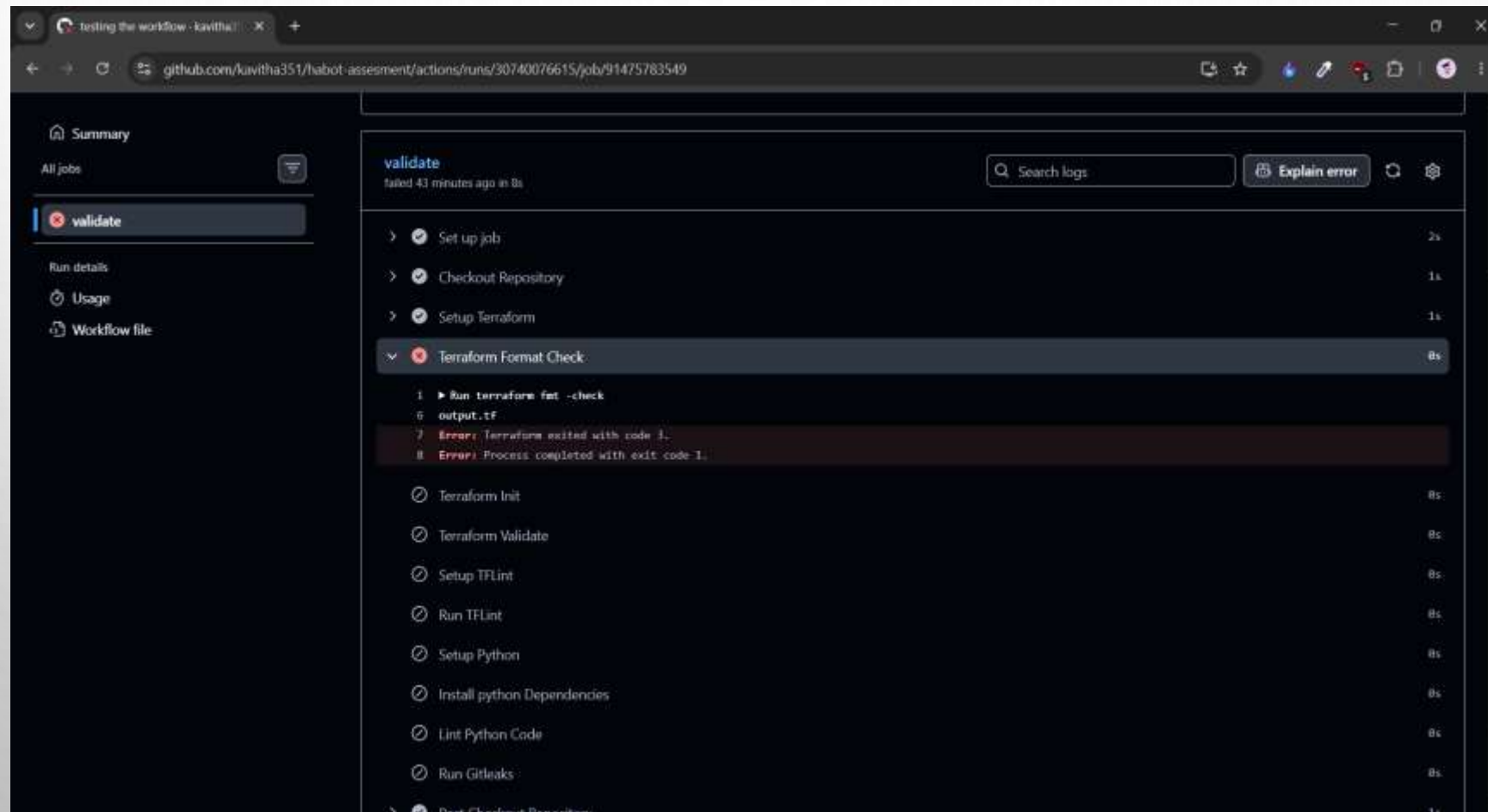
Artifacts

Produced during runtime

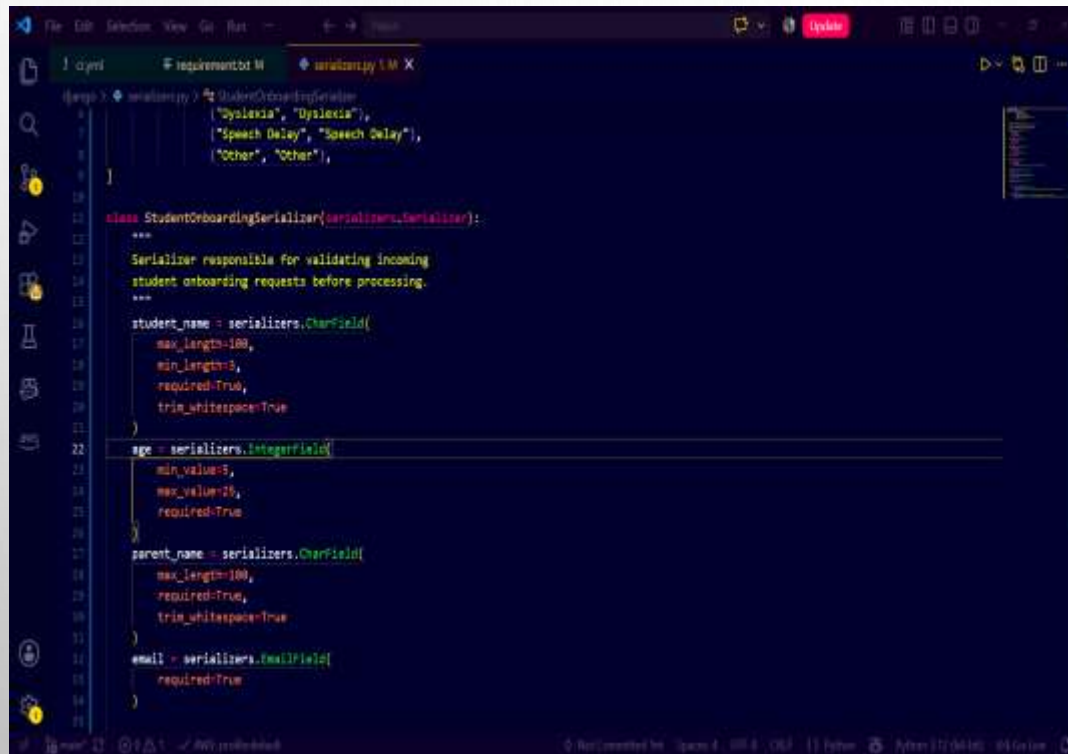
Name	Size	Digest
gitleaks-results.sarif	6.63 KB	sha256:aa1ba01018abaa75020170b9d3e8e067370dc011397... Download Delete

WORKFLOW FAIL SCENARIO

PIPELINE STOPPED AUTOMATICALLY WHEN TERRAFORM FORMATTING VALIDATION FAILED.



DJANGO



```
1 class StudentOnboardingSerializer(serializers.Serializer):
2     """
3     Serializer responsible for validating incoming
4     student onboarding requests before processing.
5     """
6     student_name = serializers.CharField(
7         max_length=100,
8         min_length=3,
9         required=True,
10        trim_whitespace=True
11    )
12    age = serializers.IntegerField(
13        min_value=3,
14        max_value=25,
15        required=True
16    )
17    parent_name = serializers.CharField(
18        max_length=100,
19        required=True,
20        trim_whitespace=True
21    )
22    email = serializers.EmailField(
23        required=True
24    )
```

- Studentonboardingserializer
- Field-level validation
- Business rule validation

DCYN (BINARY YES/NO) LOGIC LIBRARY

Input Field

Email valid?

Phone valid?

Age valid?

LSA requested?

Consent given?

Rule

Email format check

10 digits

Between 5 and 25

Boolean

Boolean

Output

Yes / No

Yes / No

Yes / No

Yes / No

Yes / No

DECISIONS

Decision

Terraform

Variables

IAM

Github Actions

Gitleaks

Serializer

Reason

Infrastructure as Code

Avoid hardcoding

Least Privilege

Automated validation

Secret detection

Strict data validation

CHALLENGES

- Understanding GCP resource provisioning with Terraform
- Building a fail-closed CI/CD pipeline
- Designing strict serializer validation
- Applying IAM using least-privilege principles

OUTCOME

- Provisioned secure infrastructure using Terraform
- Implemented automated CI/CD validation
- Built reusable request validation
- Strengthened understanding of Google security and automation



THANK YOU